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Clockwise Rotation of Southwest Japan Inferred from Paleomagnetism of Miocene Rocks in Fukui Prefecture

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In recent years, several paleomagnetic studies have been carried out on Neogene rocks from the Sanin, the Setouchi, and the Kinki districts in Southwest Japan (OTOFUJI and MATSUDA, 1983, 1984; TORII, 1979, 1983; TAGAMI, 1982; HAYASHIDA and ITO, 1984). OTOFUJI *et al.* (1985) compiled these results, and discussed the process of tectonic rotation of Southwest Japan. According to them the mean declination of the rocks older than 15.5 Ma is approximately 47° E, while that of the rocks younger than 14.5 Ma shows no significant offset from the present dipole field. They concluded that Southwest Japan was rotated clockwise through 47° within a period from 15.5 Ma to 14.5 Ma in association with the opening of the Sea of Japan.

This paper describes the paleomagnetic results obtained from Miocene rocks distributed in Fukui Prefecture, the northeastern part (the Hokuriku district) of Southwest Japan, and discusses the tectonic rotation of this area.

This area was geologically mapped in detail, and its stratigraphic relations were well established by TSUKANO and MIURA (1954), MIURA (1979), and NAKAGAWA *et al.* (1985). Furthermore, chronological studies using fission-track method (HIROOKA *et al.*, 1972; NAKAJIMA *et al.*, 1983; NAKAJIMA and MIZUSHIMA, 1984; NAKAJIMA *et al.*, 1985), which yielded ten radiometric ages ranging from 27 Ma to 11 Ma, clarified the history of igneous activity in this area. Rock samples were collected from the Nishitani rhyolites (3 sites), the Ito-o Formation (23 sites), the Uchiura Group (9 sites) and the Kunimi Formation (17 sites) in ascending stratigraphic order, and these rock types are listed in Table 1 together with results of paleomagnetic measurements. The Nishitani rhyolites, the Ito-o and the Kunimi Formations are in the Niu Mountains, and the Uchiura Group is in the Uchiura area. Figure 1 shows the sampling localities. The stratigraphies and the fission-track ages of the formations are summarized in Fig. 2. The paleomagnetic results from the Ito-o Formation and the Uchiura Group have already been published by HIROOKA *et al.* (1972) and NAKAJIMA *et al.* (1985), respectively.

Remanent magnetization was measured with an astatic magnetometer at Fukui University and a Schonstedt SSM-1A spinner magnetometer at Toyama University. Magnetic stabilities of all samples were tested by a progressive demagnetization method in alternating field (AF) in steps of 50 Oe up to a maximum field of 500 Oe. Median destructive field (MDF) values of rhyolite samples exceed 400 Oe, and those of other rock samples are between 75 Oe and 400 Oe (Table 1). Results of the AF demagnetization of typical sites (KY2: andesite samples. NF1: rhyolite samples) are illustrated in Fig. 3. No remarkable change is observed in magnetic direction although

Table 1. Paleomagnetic results from Miocene rocks distributed in Fukui Prefecture.

Site	ODF (Oe)	MDF (Oe)	N	D (°E)	I (°down)	(Tilt correction) Strike (°)	Dip (°)	Dc (°E)	Ic (°down)	α_{95} (°)	k	VGP Lat (°N)	VGP Lon (°E)	Rock type
KUNIMI FORMATION														
Ichinose Member (1)														
KY 5	50	100	9	333.0	59.1	N60E	18N	332.1	41.1	13.4	15.7	62.9	26.2	Dacite
KY 4	50	100	8	307.3	54.6	N45W	28N	343.7	49.2	6.2	81.5	75.1	26.9	Dacite
KY11	50	100	7	16.9	5.1	N33E	30S	21.5	12.4	14.6	18.0	54.2	277.6	Tuff
KY3A	300	150	4	3.3	43.8	N63E	45N	354.4	2.8	6.4	205.1	79.1	299.9	Tuff
KY3B	300	150	7	68.5	-58.6	N63E	45N	113.9	-34.6	9.0	46.3	-8.3	269.5	Tuff
OK 1	50	75	9	322.5	76.1	N40W	25N	20.2	61.1	9.1	32.9	73.3	198.9	Andesite
FM15	100	100	8	93.1	-50.7	N45W	15N	111.2	-58.9	6.3	77.5	-36.9	252.2	Dacite
Mean (1)				Dcm=342.1°E,		Icm=42.2°,		α_{95}=28.2°,		N=7				
				(Dm=327.1°E,		Im=42.2°,		α_{95}=28.8°,		N=7)				
Tohbu Member (2)														
OK 2	50	75	5	301.0	65.8	N60W	15N	330.6	61.5	13.5	33.1	66.5	71.3	Andesite
KY10	150	150	10	77.5	56.5	N45W	22N	66.7	36.8	4.2	131.9	30.3	221.8	Andesite
Ohniu Member (3)														
OK 9	100	100	9	359.8	45.8	N31E	15N	349.2	36.8	3.3	246.4	71.8	350.3	Tuff
OK10	200	75	11	33.1	49.2	N77E	15N	23.6	37.9	11.6	16.5	67.4	255.4	Tuff
OK11	100	100	10	3.6	45.4	N48E	10N	357.6	38.0	3.8	163.9	79.2	242.0	Tuff
KY 2	100	75	10	22.7	56.5	N25E	9N	9.5	55.1	4.0	144.5	82.3	228.3	Tuff
Mean (2)+(3)				Dcm=11.5°E,		Icm=48.4°,		α_{95}=21.8°,		N=6				
				(Dm=16.0°E,		Im=58.7°,		α_{95}=20.3°,		N=6)				
Fukatani Member (4)														
OK 5	200	200	6	247.5	-61.2	N63E	40N	201.0	-56.4	28.6	6.4	-20.5	8.0	Andesite
OK 6	200	150	5	261.9	-44.8	N73E	30N	232.8	-41.7	22.9	12.1	-21.1	28.7	Andesite
OK 7	100	200	7	220.5	-51.8	N53E	20N	199.8	-43.8	27.2	5.9	-56.6	40.3	Andesite
FM12	100	100	6	7.9	60.4	N53E	20N	352.1	44.2	9.3	52.4	77.7	351.7	Tuff
Mean (4)				Dcm=21.7°E,		Icm=48.7°,		α_{95}=21.5°,		N=4				
				(Dm=58.2°E,		Im=59.0°,		α_{95}=26.2°,		N=4)				

UCHIURA GROUP (NAKAJIMA *et al.*, 1985)

Quartz diolite (5)

FO 5	50	100	9	14.7	37.7	N90E	15N	12.6	23.1	13.5	15.5	63.9	286.5	Quartz diolite

Imadobana Formation (6)

FO19	150	250	5	124.9	-59.5	N85W	8N	135.1	-54.9	3.8	399.3	-53.8	238.8	Andesite
FO 6	50	200	9	42.5	58.8	N90E	15N	30.7	46.7	9.2	32.6	62.9	233.1	Andesite
FO 9	50	300	10	347.1	63.2	N35W	20N	13.0	51.3	5.9	67.5	78.6	240.1	Andesite
FO27	100	400	10	357.9	53.1	N40W	10N	6.7	46.3	5.6	74.3	80.2	277.9	Andesite

Mean(5)+(6) Dcm=6.5°E, Icm=47.0°, α_{95}=21.5°, N=5
(Dm=0.3°E, Im=58.4°, α_{95}=20.8°, N=5)

Shimo Formation (7)

FO21	100	200	10	210.7	-53.9	N 7E	14N	190.6	-57.2	11.8	17.7	-81.2	27.4	Tuff
FO13 *	500	400	9	352.8	-39.3	N80E	16N	353.8	-55.3	16.4	10.8	18.4	320.8	Tuff
FO14	50	100	8	336.0	49.4	N80E	12N	338.5	37.7	7.0	64.4	66.3	13.9	Tuff
FO15	200	400	7	3.6	33.2	N80E	12S	6.1	44.8	12.2	25.6	79.5	284.0	Andesite

Mean (7) Dcm=356.7°E, Icm=47.5°, α_{95}=24.2°, N=3
(Dm=2.6°E, Im=47.4°, α_{95}=32.7°, N=3)

ITO-O FORMATION (HIROOKA *et al.*, 1972)

Umera Member (8)

NB 5	200		12	215.2	-34.1	N20E	70N	170.5	-23.3	4.1	114.9	-64.7	158.2	Tuff
NB 4	200		7	274.6	45.8	N40W	50N	345.2	57.2	9.5	41.0	78.0	59.2	Tuff
NB 3			11	332.2	86.3	N87W	40N	0.2	46.8	4.9	86.5	82.1	314.7	Tuff

Mean (8) Dcm=352.3°E, Icm=42.6°, α_{95}=28.2°, N=3
(Dm=342.6°E, Im=70.0°, α_{95}=80.2°, N=3)

Nunogataki Member (9)

NO 1	400		8	63.2	42.8	N10E	12E	68.4	32.9	10.4	29.4	27.7	223.3	Andesite
NO 2			6	24.1	50.5	N28E	15W	7.1	47.2	2.9	536.8	80.3	276.0	Andesite
NA 1			9	52.9	49.8	N50E	50S	93.2	27.8	6.5	63.0	6.1	217.7	Andesite
NN 1			12	55.8	15.8	N40E	30N	45.6	21.5	5.3	68.2	42.0	245.4	Andesite
NR 5			12	13.8	36.3	N20W	15E	21.1	27.1	3.4	162.1	61.3	269.5	Andesite
NN 2	200		10	298.0	-22.7	N48E	35N	285.2	-54.4	5.7	73.2	-9.4	9.3	Andesite
NR 3			6	25.1	63.2	N70E	30S	98.1	68.8	12.1	31.8	23.2	177.4	Andesite

Mean (9) Dcm=58.8°E, Icm=45.3°, α_{95}=25.8°, N=7
(Dm=52.7°E, Im=45.3°, α_{95}=24.9°, N=7)

Table 1. (continued).

Site	ODF (Oe)	MDF (Oe)	*N*	*D* (°E)	*I* (°down)	(Tilt correction) Strike (°)	Dip (°)	*Dc* (°E)	*Ic* (°down)	α_{95} (°)	*k*	VGP Lat (°N)	VGP Lon (°E)	Rock type
Yokoyama Member (10)														
NY 3			*11*	52.2	52.1	N60E	50W	12.8	26.3	6.2	55.7	65.0	285.5	Andesite
NY 2			*11*	19.7	48.7	N44E	20N	3.5	37.8	9.3	24.9	74.9	303.5	Andesite
NY 1	400		*4*	188.9	8.5	N68E	36S	191.3	-22.3	7.9	136.5	-63.6	110.6	Andesite
NY 6			*3*	21.2	46.0	N68E	36S	69.6	61.6	21.5	33.9	37.3	196.0	Andesite
NY 5			*9*	348.5	51.8	N70E	30S	13.8	80.5	3.6	206.0	53.8	143.5	Andesite
Mean (10)				*Dcm*=17.5°E,	*Icm*=47.3°,	α_{95}=29.3°,	*N*=5							
				(*Dm*=17.0°E,	*Im*=40.6°,	α_{95}=30.4°,	*N*=5)							
Sasakawa Member (11)														
NM 3			*6*	346.6	59.0	N36W	12N	2.0	52.9	4.0	286.4	87.2	285.0	Andesite
NM 4			*9*	58.2	54.2	N45W	20N	54.8	34.6	2.4	446.6	39.2	230.4	Andesite
NS 5 *	200		*7*	130.7	-18.8	N70W	50S	107.9	-27.6	8.0	58.6	-22.9	227.9	Andesite
NS 2	200		*5*	199.0	-31.0	N10E	30E	213.8	-22.3	9.5	65.5	-51.0	76.2	Andesite
NS 1			*4*	355.2	55.2	N10W	50E	39.3	29.2	4.8	366.0	49.6	245.9	Andesite
NB 1			*5*	122.9	27.7	N75E	40N	94.0	51.1	5.1	225.2	15.1	197.4	Andesite
NS 6			*7*	358.7	45.3	N70E	30S	25.5	71.6	5.2	136.5	63.3	168.2	Andesite
NS 3			*10*	357.3	53.7	N70E	30S	43.8	78.7	3.8	165.2	49.6	159.4	Andesite
Mean (11)				*Dcm*=42.5°E,	*Icm*=51.4°,	α_{95}=21.6°,	*N*=7							
				(*Dm*=23.7°E,	*Im*=56.7°,	α_{95}=27.5°,	*N*=7)							
NISHITANI RHYOLITES (12)														
FM18	100	400	*9*	78.3	70.5	N74W	28N	43.0	49.4	8.0	42.8	53.8	222.4	Rhyolite
NF 1	300	>500	*11*	36.2	41.4	N65E	37S	74.3	48.2	2.3	394.4	28.5	209.4	Rhyolite
NH10	100	>500	*13*	83.5	50.0	N56E	27N	47.6	54.8	3.0	188.6	51.7	212.9	Rhyolite
Mean (12)				*Dcm*=55.3°E,	*Icm*=51.6°,	α_{95}=17.4°,	*N*=3							
				(*Dm*=62.1°E,	*Im*=56.1°,	α_{95}=34.4°,	*N*=3)							

ODF: optimum demagnetization field (1 Oe=0.1 mT). MDF: median destructive field. *N*: number of samples. *D* (*Dc*): declination of the site mean (*Dc*: after tilt correction). *I* (*Ic*): inclination of the site mean (*Ic*: after tilt correction). *Dm* (*Dcm*) and *Im* (*Icm*): declination and inclination of the stratigraphic unit mean (*Dcm*, *Icm*: after tilt correction) (*: excluded from the calculation). α_{95} and *k*: Fisher's statistic parameters. VGP: virtual geomagnetic pole position.

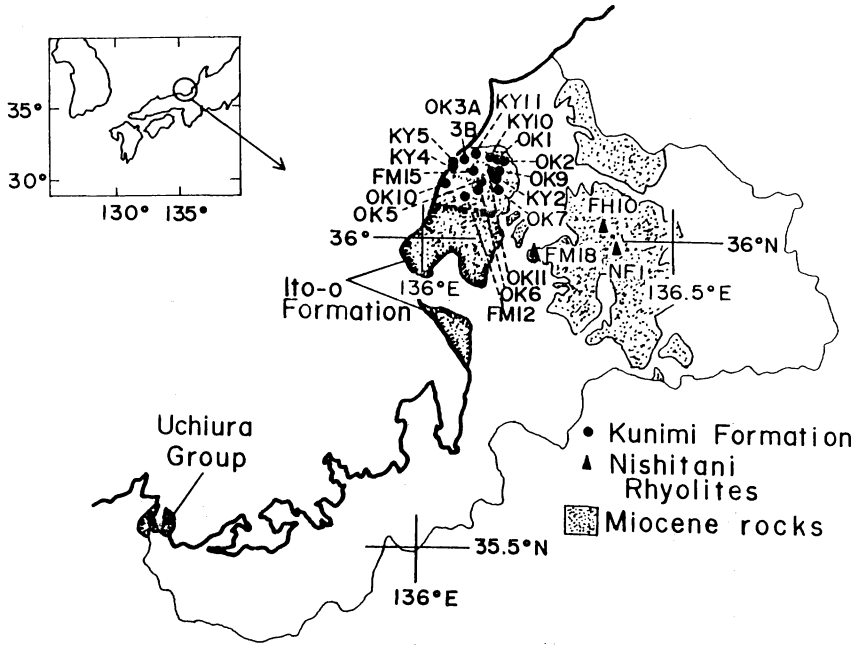


Fig. 1. Investigated area. Dots and triangles show sampling localities. The areas of the Ito-o Formation and the Uchiura Group are referred to HIROOKA *et al.* (1972) and NAKAJIMA *et al.* (1985), respectively.

MDF values are different among sites or rock types as can be seen in the figure. Site mean directions listed in columns of D (declination) and I (inclination) of Table 1 were selected from the AF treatment at the optimum demagnetization field (ODF) where the radius of the circle of Fisher's 95% confidence (α_{95}) was the smallest. Every site mean direction was corrected for the tilt of the stratum. Since the paleomagnetic results of the Ito-o Formation by HIROOKA *et al.* (1972) had not been corrected for the tilts of strata, we newly observed the tilts of the sampling sites, and corrected them. The corrected declinations and inclinations are shown in columns Dc and Ic of Table 1. A mean direction calculated for each stratigraphic unit (numbered from 1 to 12) is also listed in Table 1. In the calculation, the data obtained from the unit that has less than three sites were combined with those from the nearest unit judged from the stratigraphical view point, and reversed polarity directions were inverted to normal ones. Magnetic directions of Site FO13 of Unit 7 and site NS5 of Unit 11 are anomalous, compared with those of other sites in each unit. Each of these two units has both normal and reversed polarity directions so that at least one geomagnetic field transition occurred during the period in which the unit was formed. Since those anomalous directions seem to be attributed to the short time phenomena of the geomagnetic field such as the transition of reversals or excursions, they were excluded from the statistic computation. Figure 4 represents Schmidt's stereographic projections of the unit mean directions and their α_{95} values after and before the tilt corrections. It is apparently observed that originally divergent directions converge upon applying the tilt corrections.

NIU MOUNTAINS

	Paleomagnetic Samples
KUNIMI FORMATION (a)	
(1) Ichinose tuff Member	KY5, KY4, KY11, OK3A,
11.1 ± 1.5 Ma (f)	OK3B, OK1, FM15.
(2) Tohbu tuffaceous shale Member	OK2, KY10.
(3) Ohniu tuffaceous sandstone and shale alternation Member	OK9, OK10, OK11, KY2.
(4) Fukatani tuffaceous sandstone and shale Member	OK5, OK6, OK7, FM12.
14.1 ± 2.0 Ma (e)	
ITO-O FORMATION (b)	
(8) Umeura volcanic Member	NB5, NB4, NB3.
15.4 ± 2.0 Ma (e)	
(9) Nunogataki volcanic Member	NO1, NO2, NA1, NN1,
16.0 ± 1.8 Ma (d)	NR5, NN2, NR3.
(10) Yokoyama volcanic Member	NY3, NY2, NY1, NY6. NY5.
(11) Sasakawa volcanic Member	NM3, NM4, NS5, NS2,
18.6 ± 1.7 Ma (d)	NS1, NB1, NS6, NS3.
19.4 ± 1.0 Ma (e)	
(12) NISHITANI RHYOLITES (b)	FM18, NF1, FH10.
21.7 ± 1.7 Ma (e)	
24.5 ± 0.6 Ma (e)	
27.2 ± 0.5 Ma (e)	

Stratigraphy

(a) TSUKANO & MIURA, 1954.
(b) MIURA, 1979.
(c) NAKAGAWA et al., 1985.

UCHIURA AREA

	Paleomagnetic Samples
UCHIURA GROUP (c)	
(5) Quartz diolite	FO5.
(6) IMADOBANA FORMATION	FO19, FO6, FO9, FO27.
(7) SHIMO FORMATION	
Kohnoura shale Member	FO21.
15.2 ± 2.3 Ma (g)	
Najima volcanic Member	FO13, FO14, FO15.

Fission-track age

(d) HIROOKA et al., 1972.
(e) NAKAJIMA et al., 1983.
(f) NAKAJIMA & MIZUSHIMA, 1984.
(g) NAKAJIMA et al., 1985.

Fig. 2. Stratigraphic sequence and fission-track ages of Miocene rocks in Fukui Prefecture.

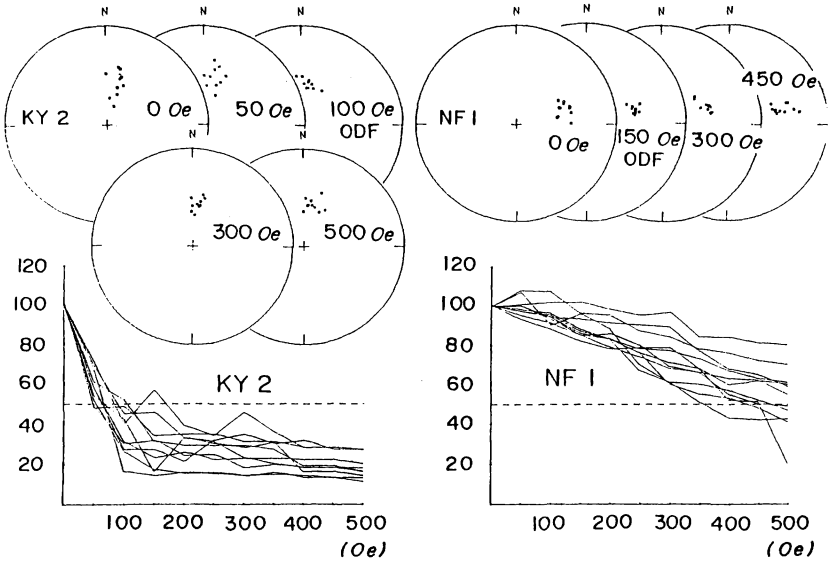


Fig. 3. AF demagnetization plots showing changes in direction and decay of normalized remanent intensity. Projections are equal area. Rock types of KY2 and NF1 are andesite and rhyolite, respectively.

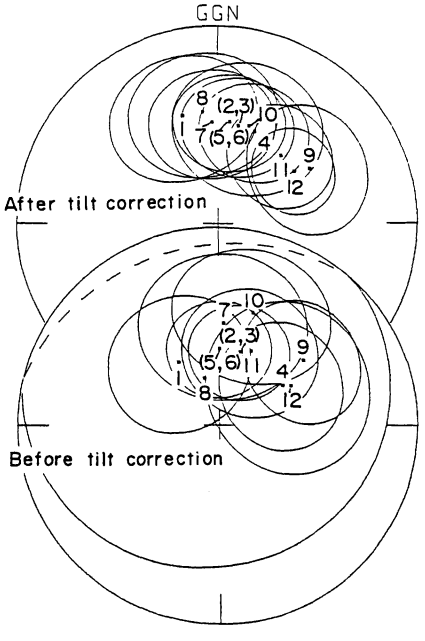


Fig. 4. Unit mean directions with α_{95} values after and before the tilt corrections. Projections are equal area. The number corresponds to that of the stratigraphic unit in Table 1.

The change in the unit mean declination (*Dcm*) with age is shown in Fig. 5. The declination error bar corresponds to the α_{95} value, and the age error bar to the uncertainty estimated from the fission-track ages and the stratigraphy. The following three features can be seen in the figure; 1) Declinations (*Dcm*) obtained in the stage before about 16 Ma deflect eastward by more than 40° (except Unit 10). 2) An abrupt declination change toward the north occurs at the point of about 16 Ma. 3) Declinations (*Dcm*) point to the north with scattering less than 22° in the stage after about 15 Ma. The mean directions of older units from Unit 12 to Unit 9 (Mean A) and of younger units from Unit 8 to Unit 1 (Mean B) were calculated separately, and are shown by dashed lines in Fig. 5. The declination, inclination and α_{95} of Mean A are 43.6° E, 50.1° and 14.6°, respectively. Those of Mean B are 1.2° E, 46.8°, and 8.6°. A declination value of each stratigraphic unit (*Dcm*) is somewhat scattered around Means A and B. That may be caused by the effect of paleo-secular variation of the geomagnetic field because the number of sites in one unit seems insufficient to cancel it out. Such an effect, however, is probably excluded from Means A and B because the data of each stage cover a long period more than 4 million years and have a sufficient number. Thus the directions of Means A and B can be regarded as the representative paleomagnetic directions during the periods of 27–16 Ma and 15–11 Ma, respectively. Mean B agrees closely with the direction of the present geomagnetic field, whereas Mean A deflects extremely eastward. Miocene paleomagnetic poles from Eurasia (IRVING, 1977) are almost identical with the present geographic pole. This implies that the deflected declination observed from Mean A is rather explained in terms of a local tectonic movement of the area than attributed to polar wander on a global scale.

The characteristic easterly declination of Mean A (43.6° E) and its abrupt change toward the north (Mean B: 1.3° E) are the similar features to the paleomagnetic

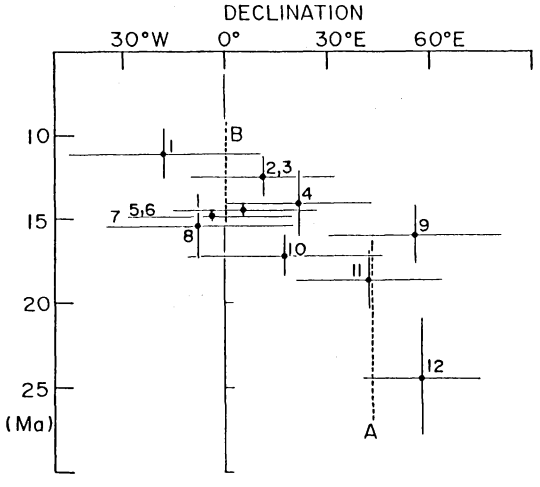


Fig. 5. Change in the unit mean declination after the tilt correction with age. The number corresponds to that of stratigraphic unit in Table 1. The declination error bar is the α_{95} value, and the age error bar is the uncertainty estimated from the fission-track ages and the stratigraphy. Dashed lines of A and B show the mean declinations of older units from Unit 12 to Unit 9 and of younger units from Unit 8 to Unit 1, respectively.

results from other areas in Southwest Japan compiled by OTOFUJI *et al.* (1985). These easterly declinations are interpreted to indicate a clockwise rotation of the land block (OTOFUJI and MATSUDA, 1983, 1984; HAYASHIDA and ITO, 1984; etc.). Consequently, the present paleomagnetic results suggest that the Fukui area was subjected to the clockwise rotation as a part of the Southwest Japan block. Based on the declination values of Means A and B, the rotation angle is estimated to be approximately 42°, which is not significantly different from the angle (47°) estimated by Otofuji *et al.*.

It is rather difficult to determine the exact timing of the rotation because of the large uncertainty of radiometric ages. As can be observed in Fig. 5, the rotational movement appears to have taken place during a very short interval between Unit 9 (16.0±1.8 Ma) and Unit 8 (15.4±2.0 Ma). The geological age of the Shimo Formation of the Uchiura Group (Unit 7), which seems to have been formed immediately after the rotation was completed, was precisely determined to be 14.93–15.23 Ma by NAKAJIMA *et al.* (1985) on the basis of the fission-track age, the micropaleontological study, and the magnetostratigraphy. Therefore, it is supposed that the rotational movement initiated at approximately 16 Ma, and ceased prior to 15 Ma at latest; i.e. the period when the rotation took place can be estimated to be 16–15 Ma. That is almost similar to the period of 15.5–14.5 Ma determined statistically by Otofuji *et al.*.

Both inclination values of Mean A (50.1±14.6°) and Mean B (46.8±8.6°) are close to that of the present local geomagnetic field (49°), but shallower than that of the axial geocentric dipole field (55°). The similar trend of the inclination was observed in the previous studies of Miocene rocks in Southwest Japan (OTOFUJI and MATSUDA, 1983, 1984; HAYASHIDA and ITO, 1984). It is difficult, however, to judge whether or not the northward translation of Southwest Japan occurred since these results have no sufficient resolution because of relatively large 95% confidence levels.

In conclusion, the Fukui area is inferred to have been rotated clockwise through about 42° within a period from about 16 Ma to 15 Ma. This area is in a northernmost part of the block of Southwest Japan, the clockwise rotation of which has been confirmed by paleomagnetic investigations.

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